

```
import java.sql.*;

public class StudentPreparedStmtApp {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/testdb?";

        String user = "testuser";

        String password = "testpass";

        try {

            // Load MySQL JDBC Driver

            Class.forName("com.mysql.cj.jdbc.Driver");

            // Connect to Database

            Connection con = DriverManager.getConnection(url, user, password);

            // a. Create Student table

            String createTable = "CREATE TABLE IF NOT EXISTS Student ("

            + "RollNo INT PRIMARY KEY, "

            + "Name VARCHAR(50), "

            + "Address VARCHAR(100))";

            con.createStatement().executeUpdate(createTable);

            System.out.println("Table created successfully.");

            // Insert initial records using Statement (for setup only)

            Statement stmt = con.createStatement();

            stmt.executeUpdate("INSERT IGNORE INTO Student VALUES (1,

            'Ravi',

            'Hyderabad')");

            stmt.executeUpdate("INSERT IGNORE INTO Student VALUES (2,

            'Sita',

            'Chennai')");
```

```
stmt.executeUpdate("&quot;INSERT IGNORE INTO Student VALUES (3,  
&#39;Kiran&#39;,,  
&#39;Bangalore&#39;)&quot;);
```

```
System.out.println("&quot;Initial records inserted.&quot;);
```

```
// b. Display content using PreparedStatement
```

```
System.out.println("&quot;\nInitial Records:&quot;);
```

```
displayRecords(con);
```

2

```
// c. Insert 2 records using PreparedStatement
```

```
String insertSQL = "&quot;INSERT INTO Student (RollNo, Name, Address) VALUES (?, ?,  
?)&quot;;
```

```
PreparedStatement insertStmt = con.prepareStatement(insertSQL);
```

```
insertStmt.setInt(1, 4);
```

```
insertStmt.setString(2, "&quot;Meena&quot;);
```

```
insertStmt.setString(3, "&quot;Pune&quot;);
```

```
insertStmt.executeUpdate();
```

```
insertStmt.setInt(1, 5);
```

```
insertStmt.setString(2, "&quot;Ramesh&quot;);
```

```
insertStmt.setString(3, "&quot;Mumbai&quot;);
```

```
insertStmt.executeUpdate();
```

```
System.out.println("&quot;Two new records inserted.&quot;);
```

```
// d. Update one record using PreparedStatement
```

```
String updateSQL = "&quot;UPDATE Student SET Address = ? WHERE RollNo = ?&quot;;
```

```
PreparedStatement updateStmt = con.prepareStatement(updateSQL);
```

```

updateStmt.setString(1, "Delhi");

updateStmt.setInt(2, 2);

updateStmt.executeUpdate();

System.out.println("One record updated.");

// e. Delete one record using PreparedStatement

String deleteSQL = "DELETE FROM Student WHERE RollNo = ?";

PreparedStatement deleteStmt = con.prepareStatement(deleteSQL);

deleteStmt.setInt(1, 3);

deleteStmt.executeUpdate();

System.out.println("One record deleted.");

// f. Display updated content using PreparedStatement

System.out.println("\nFinal Records:");

displayRecords(con);

// Close connection

con.close();

} catch (Exception e) {

e.printStackTrace();

}

}

// Function to display all records using PreparedStatement

public static void displayRecords(Connection con) throws SQLException {

String selectSQL = "SELECT * FROM Student";

PreparedStatement selectStmt = con.prepareStatement(selectSQL);

```

```
ResultSet rs = selectStmt.executeQuery();

System.out.println(""RollNo\tName\tAddress"");

while (rs.next()) {

    int roll = rs.getInt(""RollNo"");

    String name = rs.getString(""Name"");

    String address = rs.getString(""Address"");

    System.out.println(roll + ""\t" + name + ""\t" + address);

}

}

}
```